BimmerStein ECU Tool

BMW MS41, MS42 and MS43 Programming and Diagnostics

USER MANUAL

Version 0.1.0b19 - Windows

OFF-ROAD USE ONLY

Read the safety section before connecting to an ECU. Use stable power, confirm the selected operation, and never interrupt an active flash write.

Safety first

Read this section before connecting to an ECU or opening a write workflow.

IMPORTANT: OFF-ROAD, COMPETITION, RESEARCH, AND BENCH USE ONLY. Do not use this software to modify a vehicle operated on public roads. The user is responsible for compliance with applicable emissions, safety, registration, and other laws.

- Use a stable, regulated vehicle or bench power supply.
- Confirm the selected COM port, wiring mode, ECU variant, flash-chip family, and file size.
- Keep ignition ON and maintain power throughout an active write.
- Do not disconnect the FTDI adapter while a flash operation is running.
- A backup is recommended, but the Flash tab respects the operator's backup selection.
- Enable read-back verification when an independent byte comparison is required.
- Keep a suitable hardware recovery method available for boot-region or experimental firmware work.

WARNING: CODING AND TRANSMISSION CONVERSION ARE HIGHLY EXPERIMENTAL AND NOT RECOMMENDED UNTIL FURTHER TESTING. Ignition Cut and Launch Control are also **HIGHLY EXPERIMENTAL** and have not completed vehicle validation. Patches marked **Untested** remain untested.

DANGER: If a write fails after erase has started, keep ignition on and the adapter connected. Keep the application open and use **Retry Flash Recovery** while retry is available. Only when the application explicitly permits an ignition cycle, follow its instructions and choose **Confirm Ignition Cycle** to release the session and reconnect.

WARNING: Boot-region writes can replace code required for normal ECU startup. An interrupted boot write can leave the ECU unbootable and require hardware recovery. Including boot code requires explicit **WRITE BOOT** confirmation.

After a successful flash

Every successful Flash-tab write ends with the same operator instruction:

1. Turn ignition OFF.
2. Wait at least 10 seconds.
3. Turn ignition ON.

The ignition cycle is an operational handoff, not a substitute for flash finalization or optional read-back verification. After a full-ROM write, reconnect so the application detects the new program identity and available transfer routes.

Supported scope

Area	Supported scope	Important boundary
ECU families	MS41.0/.1/.2/.3, MS42 and MS43	Features follow the detected software and hardware.
Normal communication	DS2 over K-Line at 9,600 baud	Faster transfer does not change normal diagnostic speed.
Fast transfers	MS41/MS43 up to 187,500 baud; MS42 up to 166,667 baud	Settings offers a saved 125,000-baud compatibility choice.
MS42/MS43 RAM agent	Temporary high-speed reads and writes	Requires matching boot/service code; engine stopped.
MS41 Soft-BSL	Persistent loader plus RAM agents	Guided installation modifies ECU firmware.
MS41 hardware BSL	Intel 28F200 and AMD/JEDEC 29F200/29F400	Separate direct ASC0 wiring, not normal K-Line.
Full files	MS41: 256 KiB; MS42/MS43: 512 KiB	Match the family and intended operation.
Tune files	MS41: 24 KiB; MS42: 32 KiB; MS43: 64 KiB	Match the calibration to its program.
Windows packages	x64/x86 and separate Windows 7 x64/x86 packages	Choose the correct operating system and architecture.

MS42 application revisions: **0110C6, 0110C7 and 0110CA**. MS43: **430037, 430055, 430064, 430065, 430066, 430069 and 43X001**. RAM transfer also checks the boot/service code. Recognized boot variants are based on MS42 0110C6 and MS43 430055/430069; unknown boots use stock DS2 where available. Revision support does not establish physical testing of every ECU.

MS41 flash-chip families

- **Intel 28F200** uses the Intel command set. Hardware-BSL erase/program operations require the correct external 12 V VPP/RP# supply.
- **AMD/JEDEC 29F200 and 29F400** use the AMD command set and are single-supply devices. Do not apply Intel VPP requirements to them.
- A 29F400 operation must also select the correct visible upper or lower half.

Install and start

Windows installer

1. Download x64 for 64-bit Windows or x86 for 32-bit Windows. On Windows 7 SP1, choose the separately labelled Windows 7 package.
2. Run it for a per-user installation with no administrator access required.
3. Install the driver for the intended FTDI adapter, then launch **BimmerStein ECU Tool**.

The application folder and shortcuts are named **BimmerStein ECU Tool**. An update keeps the existing installation location.

Portable Windows package

1. Verify the release ZIP checksum against its matching `.zip.sha256` file when supplied.
2. Extract the complete ZIP into a writable folder.
3. Run `BimmerStein ECU Tool.exe` from the extracted application folder.

Keep the complete extracted folder together. Application runtimes, protocol resources, patch descriptors, and RAM-agent payloads are required beside the executable.

The executable may not be code-signed. Windows can show an unknown-publisher warning. Verify the exact downloaded installer or ZIP against its corresponding `.exe.sha256` or `.zip.sha256` file, or its entry in `SHA256SUMS.txt`, before continuing.

Interface driver

Install the driver for the selected FTDI adapter. D2XX is preferred for stock native-fast DS2, Soft-BSL, and hardware BSL. When D2XX is unavailable, normal DS2 and supported hardware-BSL paths can use the standard serial fallback at their compatible rates.

Data folders

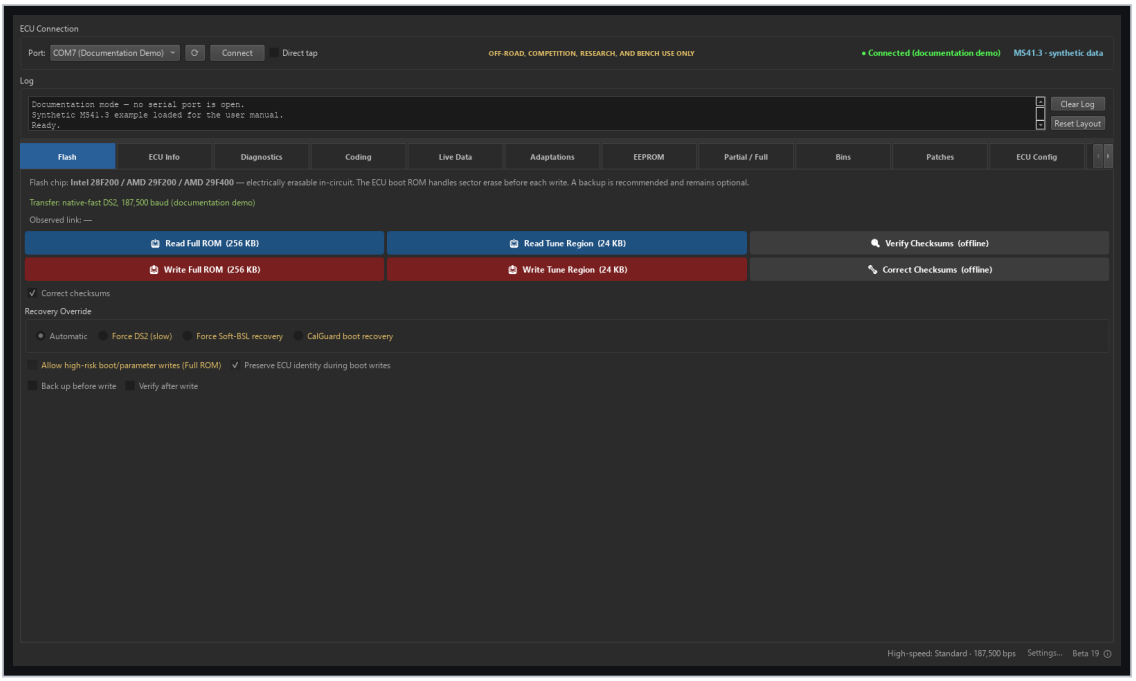
The portable application creates mutable data beside the executable:

- `backups/` for saved reads, generated images, and recovery material.
- `logs/` for session logs and diagnostic detail.

Every package includes `BimmerStein MS41 Patch Definitions.xml` beside the executable for use with a compatible calibration editor. The ROM Analyzer stores user-imported definitions under `%LOCALAPPDATA%\BimmerStein ECU Tool\definitions\`. This keeps the selected definition available when the portable application folder is replaced during an update.

Treat full ROMs and logs as private. They can contain a VIN, calibration identity, ECU identity, and operation history.

Main window



Main Flash workspace

The connection bar remains visible above all tabs. It contains the normal DS2 COM selection, Connect button, direct-tap choice, connection state, ECU variant, and transfer-mode information. The shared log and progress controls remain visible below it.

Normal **Connect** has a 30-second limit and offers **Cancel** while identifying. Cancellation or timeout restores offline controls, and the window can close even if the serial driver remains stuck. The adapter stays reserved until the pending call and cleanup finish; reconnect only when the connection control becomes available again. This applies to normal identification; active writes and retained recovery keep their existing safeguards.

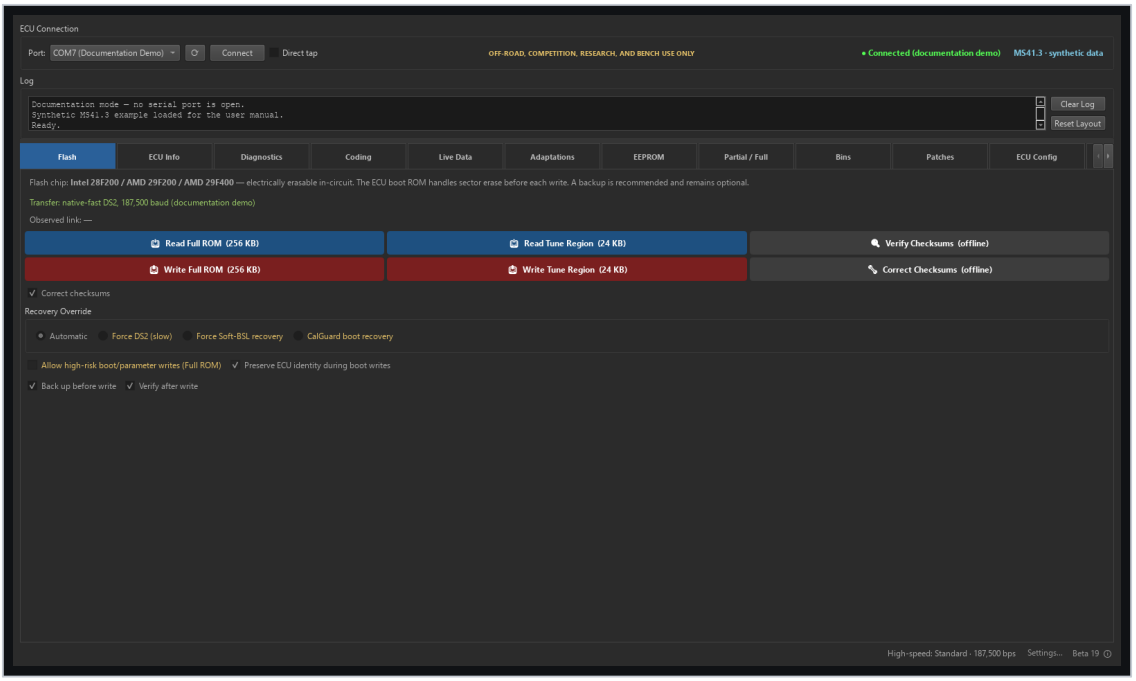
Normal K-Line versus direct tap

- Leave **Direct tap** clear for a normal single-wire K-Line or OBD-II adapter.
- Select it only for a full-duplex ASC0 connection that does not echo transmitted bytes.
- Choose the wiring mode before connecting. It is locked while a session owns the port.
- Hardware BSL has its own COM selector because it normally uses a separate adapter and wiring.

Tab order

Tab	Primary purpose
Flash	Full and tune reads/writes through the automatically selected transfer path.
ECU Info	Live identity, firmware, calibration, VIN, chip, and loader information.
Diagnostics	Scan vehicle modules, manage supported fault memories, and run controlled engine actuator tests.
Coding	Read and change recognized module settings, or run the guided transmission conversion.
Live Data	Poll supported channels for the connected ECU and record CSV data.
Adaptations	Read fuel, throttle, and knock adaptations or reset selected learned values.
EEPROM / ECU Data	MS41 EEPROM tools, or MS42/MS43 identity/history and storage backup/editing.
Partial / Full	Extract or combine calibration files using the matching full image.
Bins	Catalog reads, backups, generated images, and notes.
Patches	Compose, detect, migrate, or remove supported MS41 firmware patches.
ECU Config	Inspect and modify supported calibration configuration switches.
VIN / EWS	VIN editing and family-specific EWS alignment or synchronization.
ROM Analyzer	Inspect a BIN offline without connecting to an ECU.
Soft-BSL	Install and operate the persistent MS41 Soft-BSL loader.
BSL-Unbricker	MS41 hardware bootstrap reads, erase, program, and recovery.

Flash tab



Flash tab controls and transfer status

The Flash tab is the normal starting point for ECU reads and writes.

Choose the operation

- **Read Full** saves 256 KiB for MS41 or 512 KiB for MS42/MS43.
- **Read Tune** saves 24 KiB for MS41, 32 KiB for MS42 or 64 KiB for MS43.
- **Write Full** programs the supported full-image regions while respecting boot-write controls.
- **Write Tune** programs the calibration region.

On MS42/MS43, a stock DS2 full read is a diagnostic image: protected or reserved regions can contain FF bytes rather than the physical flash contents. Its 512 KiB file size does not make it a complete boot/identity backup. Use the RAM agent for a complete flash capture when available, and check the read source recorded in Bins. Separate EEPROM storage requires its own backup operation.

Automatic transfer selection

On MS42/MS43, Automatic mode selects the temporary RAM agent when compatible. Entry does not require a separate boot-mode read of each ECU. The agent suspends normal ECU operation and restarts it on completion, so keep the engine stopped. If entry fails, fallback to stock DS2 is possible only after a safe diagnostic state is confirmed. Force DS2 selects the stock route.

On MS41, the application selects one of these routes:

1. **Soft-BSL** when the persistent loader is detected. It starts at the highest supported tier and retries lower Soft-BSL tiers only before erase.
2. **Native-fast DS2** on a compatible stock ECU through D2XX. A failed pre-erase high-rate check can restart the complete operation over normal DS2 only after the normal ECU state is confirmed.
3. **Normal DS2 at 9600** when neither accelerated route is available.

Fallback is allowed only before erase. Once erase may have started, the application retains the active session for recovery instead of changing transports.

Write options

These controls apply to the selected ECU family. MS42/MS43 full writes normally preserve boot code and protected identity. **Include boot code** requires the RAM agent and typed **WRITE BOOT** confirmation; stock DS2 does not offer boot writes. **Preserve ECU identity during boot writes** copies live VIN/AIF, ISN/pairing and ECU identifiers into the prepared image. Separate EEPROM and retained flash storage are preserved. Boot sectors receive required readback.

The TOP/BOTTOM bank options below apply only to MS41.

- **Correct checksums** is enabled by default and corrects the final prepared image, including its boot, program, and calibration checksum fields.
- **Back up before write** is optional and follows the operator's selection.
- **Verify after write** controls host-side byte-for-byte read-back verification, with one cumulative progress indicator across the selected regions.
- ECU-side flash finalization is independent of the optional host Verify checkbox.
- For a TOP full write, **Write boot** off preserves the live 8 KiB boot region, including its identity. **Graft identity** is inactive because those bytes are already preserved.
- **Write boot** on uses the boot region from a TOP-marked file. **Graft identity** on copies the live 670 production/AIF bytes into it; off keeps the file identity.
- Soft-BSL reads only the required preservation bytes, once on the RAM-agent connection, before erase. It then writes the complete TOP image, including the coarse sector that contains the boot region. Retrying uses that same prepared image.
- BOTTOM boot/parameter writes retain their existing separate arming behavior.

Failure and retained-session recovery

Failure before erase

Before erase, supported fast routes can retry a lower rate or return to stock DS2 only after the appropriate ECU state is confirmed. MS42/MS43 boot writes cannot fall back to stock DS2 because that route does not write boot code.

Failure after erase

After erase begins, changing baud rate, reopening the port, or cycling ignition can discard the only live recovery path. The application therefore retains:

- The current adapter session for either a normal or native-fast DS2 write failure.
- The loaded RAM agent and port ownership for a Soft-BSL or MS42/MS43 RAM-agent write failure.
- The exact prepared target bytes required for a same-session retry.

MS42/MS43 retains its prepared target and active RAM session for supported readback and sector recovery. Use the retry button offered for that operation; do not change the file, rate or boot options during recovery.

MS41 first attempts the recovery supported by the active protocol:

- **DS2:** an uncertain program reply triggers a read of that packet. Matching bytes allow the write to continue. Blank bytes can be sent again, with three total packet attempts. Native-fast calibration can also continue at the first mismatching byte when the rest of that packet is proven erased. It never resends the already committed prefix.
- **Normal DS2 and native-fast full writes:** a confirmed partial packet can trigger one automatic replay of the failed erase/write phase. If calibration fails after the program phase completed, only calibration is erased and rewritten. DS2 erases firmware-defined regions, not individual packets. One further operator-requested replay is available when the retained session permits it.
- **Native-fast calibration:** another erase in the interrupted high-rate session is not supported. The app retains the connection but does not offer a re-erase that would be rejected by the ECU.
- **Soft-BSL:** up to nine identical packet transmissions handle retryable errors. An interrupted frame is drained before a CRC-protected read confirms communication. A confirmed mismatch that requires erase can replay one complete selected physical sector once, using the prepared image. One operator-requested retained replay is available if automatic recovery cannot finish.

These checks run after failures; they do not add readback to successful writes. Mandatory DS2 finalization and the selected optional Verify still run after recovery.

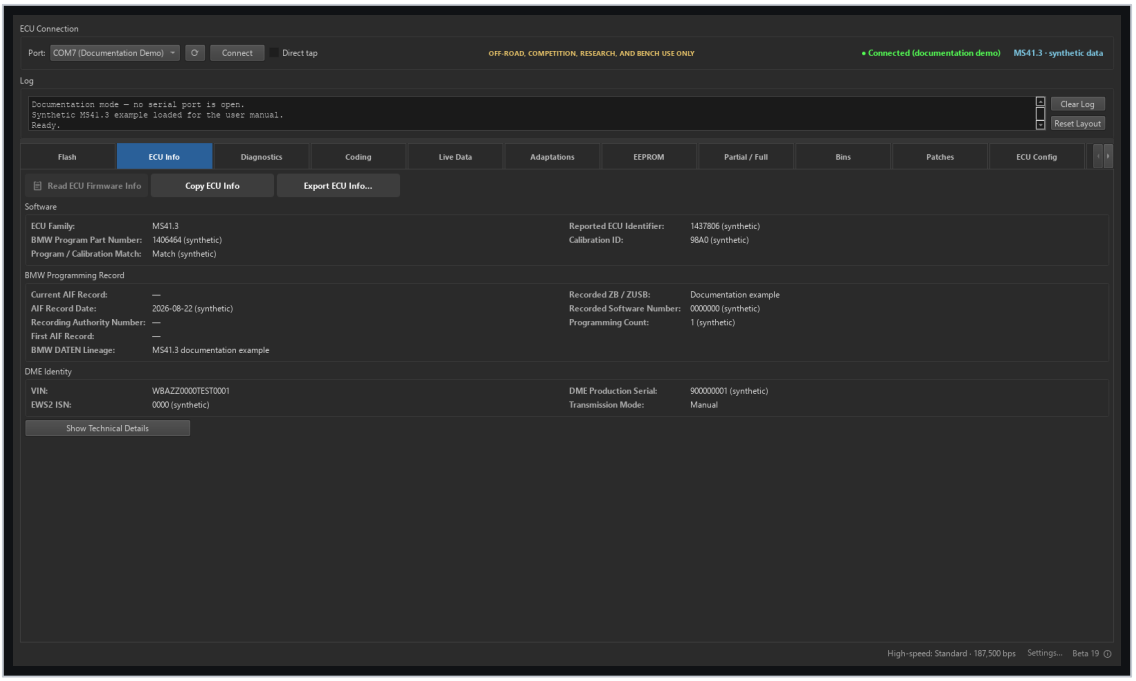
Follow the red recovery message exactly. Keep ignition ON while a session is retained. Select **Retry Flash Recovery** only when it is offered; do not independently change baud or reconnect. Use the app's controlled ignition-cycle instructions when it reports that a cycle is required.

When hardware BSL is required

For MS41, use the BSL-Unbricker workflow when normal DS2 no longer starts and no retained session can recover the ECU. MS42/MS43 hardware recovery is outside this application's BSL tab; use an appropriate separate recovery method.

DANGER: Do not experiment with reset lines, ignition cycles, VPP, or chip-family selections during a retained recovery session. Preserve the session first. For MS41, use hardware BSL only when the retained path is no longer available or cannot recover the target. MS42/MS43 requires a separate hardware recovery method if its software recovery path cannot restore operation.

ECU Info, Diagnostics, and Live Data



Synthetic ECU information workspace

ECU Info

Use the ECU Info read button after connecting. The tab can report the detected ECU variant, firmware/ECU ID, calibration ID, VIN, identity strings, flash-driver signature, flash-chip family, and Soft-BSL marker when available.

If MS41.2 and MS41.3 identification is ambiguous, use a full image or the stronger program and calibration markers rather than relying only on the shared ECU ID.

MS42/MS43 also shows ECU number, ISN and AIF programming history where available. Post-relay supply voltage and ignition supply (KL15) are distinct measurements. Bench wiring may not energize both paths; compare the appropriate circuit with your measured supply rather than treating the fields as interchangeable.

Diagnostics

Use **Faults** for module scans and fault memory. **Actuator Tests** is a separate view for the connected Engine ECU; the module selected under Faults does not change the actuator target. Both views retain the application's normal font and display scaling.

1. Leave **Direct tap** clear and connect through normal K-Line.
2. Select **Scan Vehicle** for a read-only identification scan of the six verified module addresses. Cabin OBD may expose only powertrain modules; use the BMW 20-pin diagnostic path when applicable.
3. Select the responding module. For ABS/traction-control or automatic-transmission fault memory, select the exact fitted system profile. If no embedded profile matches, fault reading and clearing remain unavailable rather than using an assumed record layout.
4. Select **Read Faults**, then review the code, reference, system, status, description, and selected-fault detail. The first fault is selected automatically; its **Freeze Frame** appears first in the detail pane, followed by fault details. Select another row to inspect it, and scroll inside the pane or drag its divider for more space. For the DME, choose **Stored** or **Shadow** memory. The same fault can appear in both; shadow memory also includes faults before they meet the ECU's logging threshold. Supported records include frequency, conditions and decoded freeze-frame values.
5. Select **Export to Text** when a service record is required.
6. Save the fault report and correct the cause before **Clear Faults**. DME clearing automatically re-reads stored memory, distinguishing remaining faults from a failed confirmation read. For other modules, read faults again to confirm clearing.

Copy ECU Info and **Export ECU Info…** in the ECU Info tab export the displayed identity fields and raw identification response with a UTC report timestamp.

Under **Actuator Tests**, select the named output directly. MS41.0 offers 12 tests; MS41.1/.2/.3 offer 17. MS42/MS43 lists its own supported outputs. Keep ignition on and the engine stopped unless the button is marked **Engine running** (VANOS and exhaust flap). The app checks RPM before starting. Output availability also depends on the vehicle's equipment and ECU configuration.

Tests run for up to five seconds after the request, or until **Stop Test** is selected. The app ends the ECU test session before enabling another operation. If the ECU does not confirm that cleanup, **Retry Stop** remains available and other ECU tasks stay blocked. Closing the window requests Stop and waits for cleanup. If communication cannot be restored, switch ECU ignition off and use the explicit Close/Disconnect confirmation to leave the unconfirmed session.

An ECU reply does not verify the physical component. Some later-family outputs can return an ambiguous result even when actuated; the app retains an explicit warning to check whether the output responded instead of reporting a passed component test.

Under **Roughness**, select **Start Monitoring** with the engine running. MS41.0/.1/.2/.3 use the roughness system check. MS42/MS43 passively reads its supported roughness values rather than starting that MS41 check. **Box plots** compares all six cylinders on a shared, symmetric scale centered on zero using the latest 30 seconds of valid readings. Each box spans the middle 50% with a line at the median. Whiskers stop at the outermost readings no more than 1.5 box widths beyond each box edge; dots show readings beyond them. A separate marker shows the current value. Each row reports its sample count; fewer than four readings shows **Collecting**. **Trend** shows the same readings over time. Switching views preserves the capture. The scale expands when needed and resets when a new monitoring session starts.

Values preserve the ECU's signed, unitless measurements: negative and positive values appear on opposite sides of zero without an absolute-value display conversion. A dash means unavailable; zero and minus one can both be valid measurements. Time labels record when the app read the values, not the ECU's internal calculation time. Missing readings leave gaps in the trend and are excluded from box statistics. Retained boxes are marked as historical when the current value is unavailable; they do not imply a current measurement.

Stop Live Data polling before starting Roughness, and stop Roughness before starting Live Data. They share the ECU connection and cannot run together. **Stop** finishes the active monitoring session, clears current values and retains the capture. If the ECU does not confirm Stop, select **Retry Stop** before another ECU operation. Closing or disconnecting waits for cleanup; if communication cannot be restored, switch ECU ignition off before accepting the explicit disconnect confirmation.

Live Data

Choose channels before **Start Polling**. The selection controls ECU reads, CSV columns, and all three views: **Values**, **Graphs**, and **Instruments**. Required validity reads may accompany selected values. Selection is locked during capture so the CSV schema stays fixed. Enabled CSV logging writes the session under the application `logs/` directory.

Favorites use stable parameter IDs and are saved separately for each exact ECU ID, program variant and logger definition. Bundled-definition favorites survive updates; imported XML favorites are tied to the file's contents. With no saved selection, the first three available channels are selected. **None** remains empty until you choose channels again.

The top row groups logger/file controls, Interval and Mode, and **Log to CSV**. **Start Polling** and **Stop** stay on the right. Open **Logger: Bundled / Imported** to inspect the current definition, **Import XML…**, or **Use Bundled**. The full definition name and path are available on hover. Change definitions while polling is stopped. The active logger supplies the supported channels, decoding and scaling; the bundled profiles include the current firmware and patch corrections. **Maximum rate** (0 ms) is the default and adds no artificial delay; the interval is saved between sessions. You can instead set 100-5000 ms. The displayed samples/s uses exact completion times from the latest three seconds, excluding initial setup. It remains zero until two samples complete and falls back toward zero if acquisition stalls or stops.

On **MS41**, Telegram batches the selected channels and their required validity data. Overlapping RAM bytes are read once; adjacent bytes are packed into aligned word reads where possible. A setup packet holds up to 49 byte/word

entries. Remaining data uses grouped RAM reads in the same cycle. Standard DS2 uses grouped RAM reads for the selection. Auto compares the plans, choosing fewer recurring transactions, then fewer bytes, and allows Standard fallback if setup fails.

On **MS42/MS43**, a configured telegram holds up to 24 supported measurement descriptors. Remaining channels use the ECU's fixed measurement groups; Standard uses those groups for the selection. Auto compares transaction counts and uses the fixed groups when they are more efficient or telegram setup fails. Telegram availability depends on the selected channels and a matching profile. The **None** button selects no channels on either family.

Actual sample rate depends on the ECU, adapter and selected channels. The displayed rate describes completed samples rather than a promised transfer rate.

The first completed sample is time zero in both the graph and CSV. Initial program detection and setup happen before that origin. With **Follow latest** enabled, graphs fill a 0-30-second window, then scroll through the latest 30 seconds. **Time +**, **Time -** and **Fit time** appear only in Graphs and change the time axis. They leave card sizes unchanged. Manual time controls turn off Follow latest; enable it again to return to the newest sample. Imported logs use their recorded range.

The bundled definition includes **Cylinder 1 Roughness** through **Cylinder 6 Roughness** for the supported MS41.0/.1/.2/.3 program IDs. When selected, these channels use ordinary RAM reads and join the same Live Data sample and CSV row as the other parameters. They never start the BMW system check. Native engine-running and validity gates leave unavailable values blank; signed values and zero are retained. An engine-stopped bench session therefore shows blank roughness, even if the Diagnostics command returns zero-filled stored words. Telegram mode includes selected roughness values and their validity data in its batch where space permits, with remaining bytes read in the same acquisition cycle. Selecting more data takes bus time and can reduce the sample rate.

Use channel search to find values before capture. Move the sample cursor, zoom or pan to inspect a capture. The live view retains the latest 5,000 samples; enabled CSV logging records the complete session. **Open CSV…** opens a saved log in the same viewer. Unavailable values remain unavailable. The single-line footer shows the capture mode, sample rate and CSV status or filename. Stopped captures show their result there; failures retain a visible **Error** prefix. Hover over the footer for complete messages, filenames and the history-retention note. ECU reads and CSV writes run in a background worker. Display updates run at up to five refreshes per second independently of acquisition; only visible graphs and instruments are painted.

In **Instruments**, choose **Gauge**, **Digital** or **Box plot** on each numeric parameter's card. Mix representations freely, including during capture. Text and on/off states stay digital. Each card keeps the same footprint and the existing channel colors; **Smaller**, **Larger** and **Fit panel** change card size only. Fit panel chooses the largest of four readable sizes that fits, including an extra-small size for more channels. Larger selections remain scrollable. The size is remembered across restarts. Per-channel display choices use the same ECU, program and logger-definition scope as channel favorites, including preserving choices for temporarily unavailable parameters.

Digital cards show the current value and min/max since capture or **Reset min/max**. Reset is available at the latest sample and leaves the capture and CSV untouched. While inspecting history, min/max uses retained samples through the cursor, after the most recent reset. Gauges keep logger-defined ranges where available; other numeric gauges explicitly show **Auto range**, based on recent samples.

Box plots summarize valid samples from the last 30 seconds through the selected cursor: the box spans the middle 50%, its line is the median, whiskers extend to the last points within 1.5 times the box width, and dots mark outliers. A white diamond marks the current sample. Until four valid samples arrive, the card shows the current point and sample count without drawing a quartile box. Missing values never become zeros. Roughness defaults to box plots; all six cylinders share a zero-centered scale. Fuel-trim bank pairs also share zero-centered scales. These displays reuse captured values and issue no additional ECU requests.

Available channels include EVAP purge duty, front-O2 and MAF input voltages, and operating states such as closed throttle, part/full load, deceleration fuel cut, and engine start. On MS41.3, the tool checks the runtime wideband flag once when polling starts. If enabled, available channels include actual AFR, target AFR, configured wideband input voltage, selected input and narrowband-emulation status.

For MS41.0, MS41.1, MS41.2, and MS41.3, Live Data identifies the installed Ignition Cut program and RAM layout once when polling starts. When recognized, it exposes standalone calibration settings, physical inputs and

independent requests. Launch Control settings and launch-arm state also require an installed Launch Control program hook. The Launch settings include clutch polarity, soft and hard cut RPM, arm and maximum speed, minimum TPS, ignition hysteresis, and fixed IPW. When the program identity is absent or unavailable, these patch-specific rows stay hidden. The runtime marker remains a separate selectable status rather than proof of installation. Current revisions also expose their timing-band and angle settings, pending recovery flags, and last published timing state/source. **Off** and invalid angle settings remain labels rather than being plotted as real angles. Published timing state describes the firmware decision; it is not measured physical spark timing. Native fuel trims, lambda state, stock-limiter state and supported heater channels remain available with stock firmware. Selected native and patch channels share the same read plan and completed CSV sample. `O2 Heater Front/Rear` values are commanded heater duty, not ignition-coil dwell. Rear-heater and raw lambda-monitor fields are omitted on firmware versions where the logger definitions do not provide a valid address.

On MS41.3, the logger also exposes native MAF-fault, manual AlphaN and Speed Density mode flags. When the current AlphaN failsafe V3 program is identified, extended engine load and MAF/model channels are available with their required load scaling. They remain unavailable in Speed Density mode or when the scaling does not match, because those RAM mirrors are not established as current values in that mode. A MAF-fault flag alone does not prove that the failsafe patch is active.

Cut-active diagnostic coverage follows the monitor paths present in each firmware: MS41.0 guards lambda regulation, upstream O2 voltage, and the shared routine containing per-cylinder roughness/misfire detection plus coil/resistor diagnostics. Its stock DTC descriptors identify coil codes 29/31/30/3/1/2 and feedback-resistor code 56, but contain no dedicated per-cylinder DTC 238-243 descriptors. The shared guard bypasses the detector and downstream diagnostic calls during an intentional cut; no synthetic DTC 238-243 path is added. MS41.1 and MS41.2 also guard rear O2, catalyst, and misfire paths; MS41.3 guards front/rear O2, catalyst, misfire, and coil/resistor paths. Offline exact-byte execution verifies the descriptors and native fault-manager release path. O2-heater electrical diagnostics remain active on every version.

Live Data is read-only with respect to flash memory.

Coding and guided transmission conversion

Vehicle module coding

HIGHLY EXPERIMENTAL - NOT RECOMMENDED UNTIL FURTHER TESTING. The same caution applies to guided transmission conversion. Keep original coding and a suitable recovery method before changing a module.

Module coding requires normal K-Line mode; Direct Tap reaches only the Engine ECU.

1. Choose the vehicle chassis and module, then select **Read Settings**.
2. The application identifies the connected coding version and exposes only an exact built-in profile. An unknown module version remains unavailable.
3. Change only understood settings. **Show advanced options** reveals reviewed equipment-specific settings and references; it does not enable raw-byte editing.
4. Select **Write Changes** and review the exact old-to-new values and battery-voltage notice.
5. Keep ignition ON and the engine OFF while writing. A successful operation reads the module back and requires the decoded settings to match. Cycle ignition before testing the change.

Guided transmission conversion

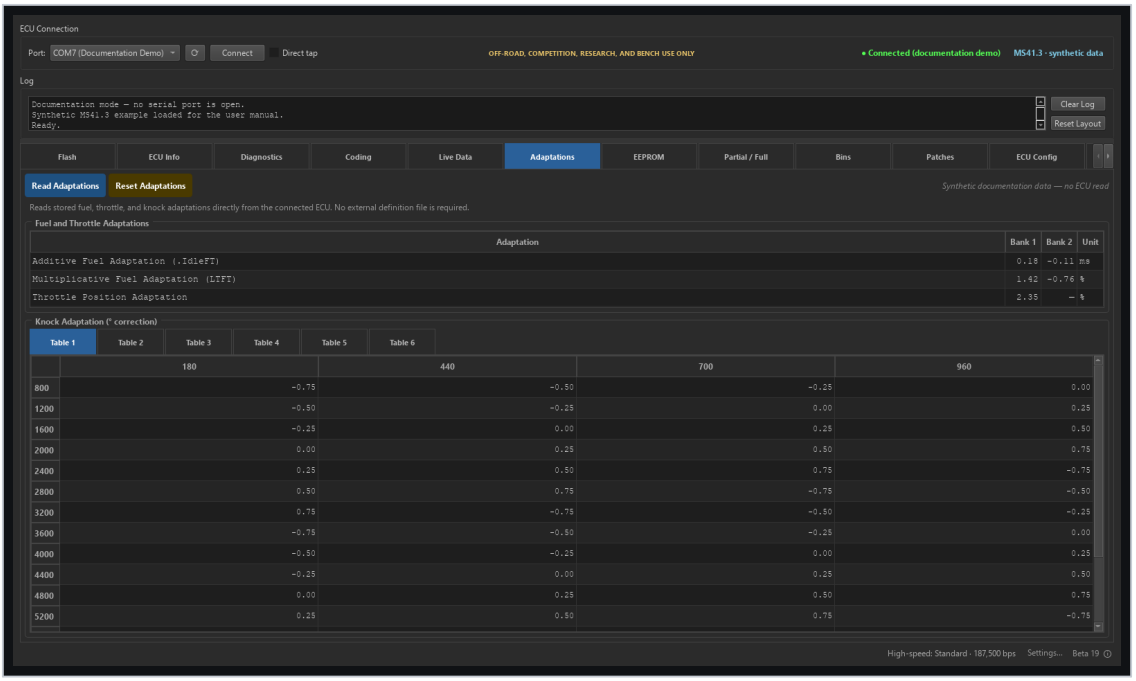
Supported exact E36/E39/E46 profiles are checked individually. Complete E53/Z3 conversion and unlisted donor combinations remain unavailable.

The compatibility check is read-only. It admits only an exact built-in chassis, engine-computer, vehicle-order, fitted-module, and transmission-computer combination; unresolved or conflicting state stops before a write.

1. Connect through normal K-Line, choose **Manual transmission** or **Automatic transmission**, and select **Check Compatibility**.
2. Review every reported owner, warning, blocking reason, and planned change. Complete the mechanical transmission and wiring work before continuing.
3. Select **I confirm the physical transmission swap is complete and the vehicle is safely parked**, then select **Convert to Manual** or **Convert to Automatic**.
4. Keep ignition ON and the engine OFF until the application explicitly requests a cycle. Before the first module write, the application durably archives the admitted before-state and starts a recovery journal. Each written owner is read back before the workflow advances.
5. When instructed, turn ignition OFF, wait at least 10 seconds, then turn ignition ON with the engine stopped.
6. Confirm the completed cycle and select **Verify Conversion**. Final verification independently reads every changed module and the applicable DME transmission record.

If a conversion is interrupted, do not start another plan. Use **Recover Interrupted Conversion...** to load the checked local record and either finish the reviewed target or restore the archived original coding. If the DME transmission-record result is unresolved, keep ignition ON and use the guided recovery; no success or rollback is assumed.

Adaptations



Adaptations tab example

For MS42/MS43, the initial **Not read** view is empty until a successful read. Only mapped learned values and supported reset groups are offered. Unknown or unavailable values are not replaced with zeros.

For MS41, **Read Adaptations** reads stored fuel, throttle, and six 16-by-4 knock-adaptation tables from the connected Engine ECU. The exact ECU ID must have packaged address and axis mappings; the operation stops when those mappings are unavailable instead of estimating them. Fuel values without an exact mapping remain shown as `-`. The knock tables use load columns, RPM rows, and degrees of correction.

Reset Adaptations changes learned ECU state and requires confirmation. MS41.0 offers **All adaptations** only. MS41.1/.2/.3 offer **All adaptations**, **Idle adaptation**, **Knock adaptation**, **Lambda / fuel trim adaptation**, or **Throttle adaptation**. The ECU relearns the selected values during subsequent operation. After reset, the tool rereads the values. If that refresh fails, it clears the old display and reports that refreshed values are unavailable.

ECU Data - MS42/MS43

The tab changes from **EEPROM** to **ECU Data** for MS42/MS43. MS41's EEPROM controls remain available when working with MS41.

Read Identity / History reads identification, VIN, ISN and programming history through normal diagnostics. **Load Report** and **Save Report** open and save that report. This operation does not read the whole storage memory.

The memory selector has separate storage choices:

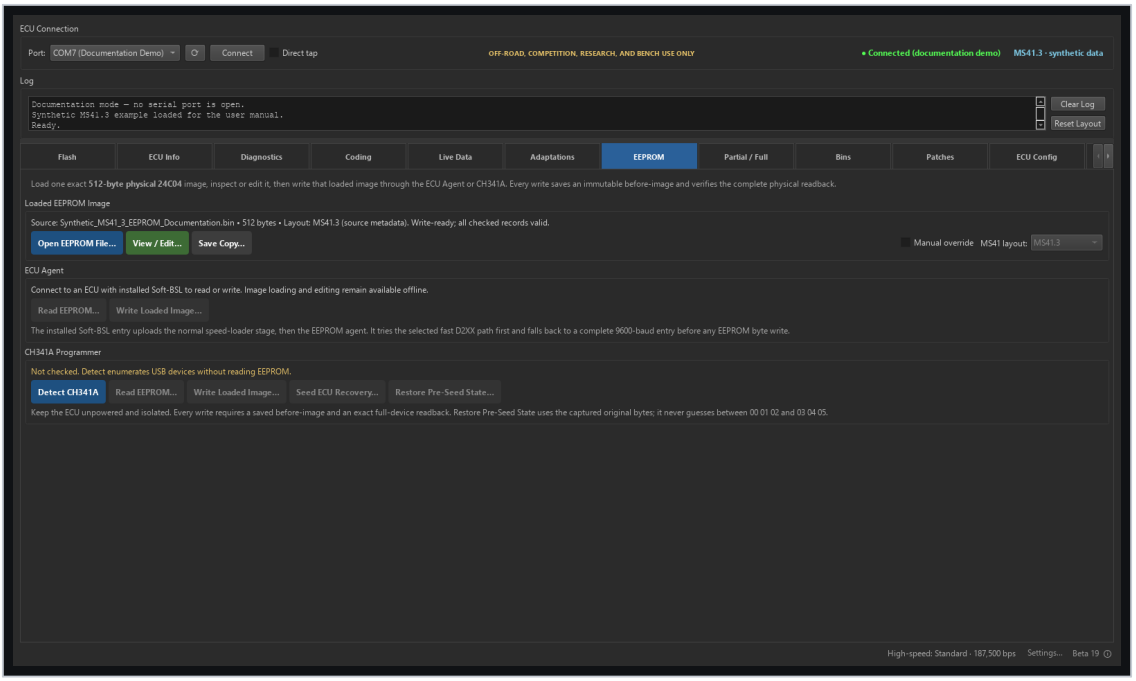
ECU	Memory	Purpose
MS42	Flash journal, 16 KiB	Retained learned values and diagnostic records
MS43	EEPROM, 2 KiB	Separate serial EEPROM, stored low byte first
MS43	Flash banks, 16 KiB	Raw storage whose operational use remains unverified

Read Flash Journal, **Read EEPROM** or **Read Flash Banks** uses the matching RAM reader when available. Flash reads may fall back to stock DS2; MS43 EEPROM does not have a stock DS2 read fallback. A flash-bank backup containing FF bytes is not an EEPROM backup.

Inspect / Edit Backup shows decoded fields, checks and raw bytes. Established values have units and names; unknown fields stay raw or read-only. Matching calibration is needed for calibration-dependent fields and table axes. The editor supports undo/redo, comparisons and explicit selected-check repair. Mouse-wheel scrolling does not change values.

Apply changes to the loaded backup, then use **Save Loaded Backup** to keep them. **Restore Loaded Backup** is a separate confirmed write with readback. If interrupted, use **Retry Storage Restore** while the session remains available. Ordinary flashing and identity grafting do not run this restore. The selected memory and loaded backup are labelled independently.

EEPROM - MS41



EEPROM workspace

The EEPROM tab works with one displayed 512-byte physical 24C04 image. Loading a file or an `EEPROM` entry from Bins replaces that same target; neither action writes to hardware.

DANGER: Do not write a short live RAM mirror padded to 512 bytes, a uniform failed capture, or an image whose exact MS41 layout is unknown. EEPROM data can contain vehicle identity and coding; keep captures private.

Load and inspect an image

1. Select **Open EEPROM File...**, or select an `EEPROM` entry in **Bins** and choose **Open in > EEPROM**.
2. Confirm the detected MS41 layout. If it is unresolved, enable **Manual override** and select the exact MS41.0, MS41.1, MS41.2, or MS41.3 program layout before editing or writing.
3. Select **View / Edit...**. The **Decoded** view provides categories, search, typed edits and learned-knock cells. Advanced edits require an explicit unlock. **Hex** retains expert byte editing, disabled by default. Undo and Redo apply to the current draft.
4. Changed-record checks use the shared EEPROM owner. **Repair selected checks** is a separate, confirmed action for selected invalid records; repair does not validate their payload. **Compare file…** shows named and raw changes. An optional matching BIN supplies verified knock-grid axes through the active calibration definition.
5. Select **Apply Changes**, review the differences, and confirm to return the edited image to the EEPROM tab. This updates the displayed target only; it does not write to hardware.
6. Use **Save Copy...** when an additional offline copy is required.

ECU Agent

The ECU Agent requires a connected, recognized MS41.0-MS41.3 program with its matching installed Soft-BSL entry and live identity evidence.

1. Select **Read EEPROM...**. A successful stable 512-byte capture is archived in Bins before an additional external copy is offered.
2. Load or edit the exact target and verify that its selected layout matches the connected program.
3. Select **Write Loaded Image...**, review the source and layout, then type `WRITE EEPROM`.
4. The writer saves an immutable before-image, compares before writing, writes only changed bytes with checked-record checks last, and requires an exact full-device readback. If the EEPROM already matches, no write is sent.

If a write becomes uncertain, keep ignition ON and do not disconnect the adapter. Select **Resolve Pending Write...**. The retained session first queries the current state; if remaining prepared operations are required, it asks for `RESUME EEPROM`. After resolution, perform and save a fresh full read before another change.

CH341A Programmer

Disconnect the normal ECU session and keep the ECU unpowered and isolated.

1. Select **Detect CH341A**. Detection enumerates USB devices; it does not read EEPROM.
2. Select **Read EEPROM...** to obtain and archive a stable full-device capture.
3. Load or edit the intended target and confirm its exact layout.
4. Select **Write Loaded Image...** and type `WRITE EEPROM`. The same programmer session performs a stable pre-read, saves the before-image, writes only changed bytes, and verifies the complete target.
5. **Seed ECU Recovery...** is available only for the displayed verified capture when its marker is eligible. **Restore Pre-Seed State...** uses only the exact original bytes captured by that seed operation; it never guesses them.

After reconnecting the programmer or restarting the app, detect and read the EEPROM again. Restore becomes available only when the exact saved original is found; missing or conflicting backups stay blocked.

After an uncertain CH341A write result, perform and save a fresh full read before any further mutation.

Offline files, conversion, and Bins

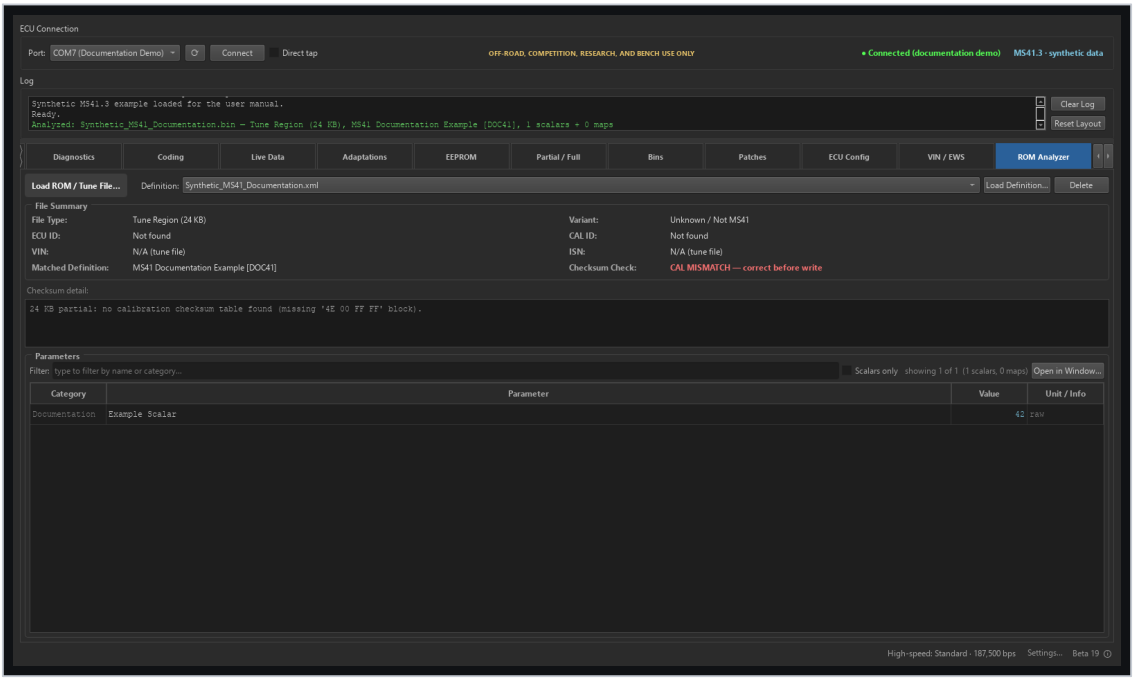
ROM Analyzer

Load a supported full or calibration BIN to inspect its size, ECU/software identity, VIN, checksums and matching definition information. No ECU connection is required. MS42/MS43 uses matching XDF definitions; MS41 uses its XML format.

To enable parameter matching:

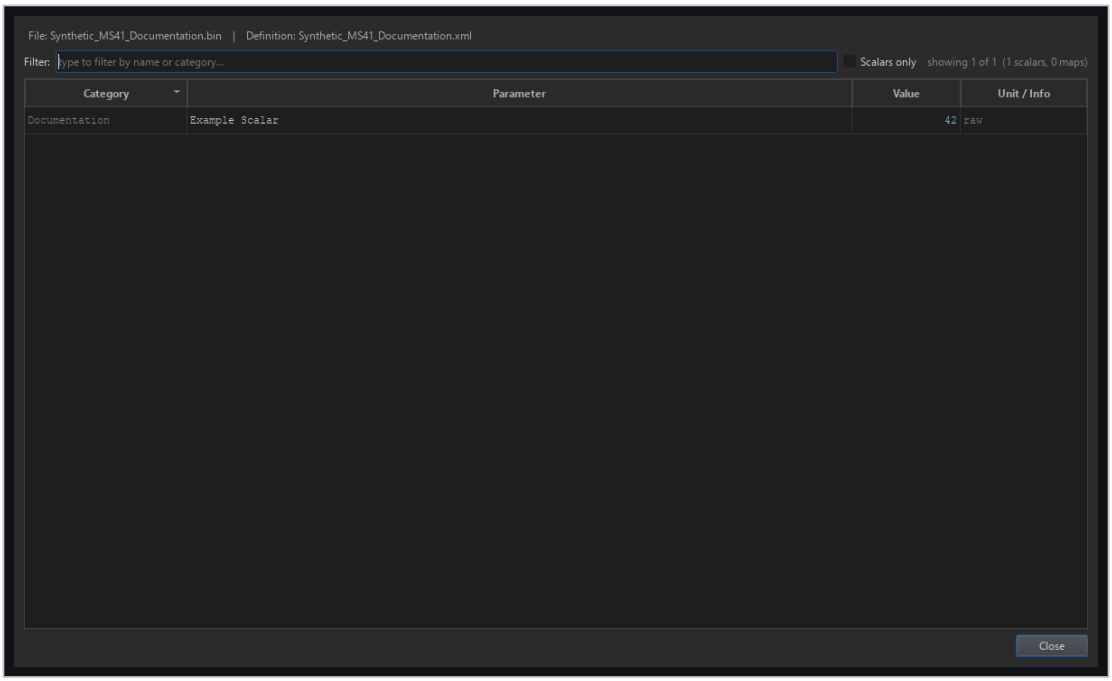
1. Select **Load Definition...** and choose a matching MS41 XML or MS42/MS43 XDF definition.
2. The tool validates the definition and copies it into the per-user registry.
3. Use the **Definition** list to switch between registered definitions. The selection persists across application restarts.
4. Select **Delete** to remove only the registered copy. The original definition is never modified.

If an imported filename already exists with identical content, the existing copy is selected. If the content differs, the tool asks before replacing the registered copy. Flashing and dumping do not depend on ROM Analyzer definitions.



ROM Analyzer example

Select **Open in Window...** in the Parameters section for a larger, resizable table. The detached window remains modeless, has its own filter and scalar-only control, supports column sorting, and updates automatically when the loaded BIN or selected definition changes.



Detached ROM Analyzer parameters window

Stop if the analyzer reports a hybrid program/calibration combination. A hybrid image can contain code and calibration from different MS41 variants and is blocked from normal flashing.

Partial / Full

- **Full to Partial** extracts 24 KiB from MS41, 32 KiB from MS42 or 64 KiB from MS43, using the matching family layout.
- **Partial into Full** replaces the calibration in a selected full base.
- Variant conversions preserve or replace identity only according to the explicit workflow. The same conversion policy applies to MS41.0, MS41.1, MS41.2, and MS41.3.

Bins

Bins catalogs files created by reads, backups, and patch composition. Entries include available variant, type, VIN/CAL metadata, notes, and source. Use descriptive notes and preserve a known-good original separately from edited or patched images. Newly cataloged files also record a SHA-256 identity so an externally replaced or modified file can be identified later.

The compact toolbar keeps Import, ECU Backup, Compare and Flash visible. **Open in** groups Tuning Suite, BSL-Unbricker, Patches, ECU Config and EEPROM destinations. Use the folder filter and **Organize > New folder…**, **Rename image…** and **Move to folder…** to organize images. Folders are real directories under backups. Moves and renames preserve metadata when the file match is unambiguous; conflicting filenames are not overwritten. Browse > Refresh also discovers supported files managed outside the app. Under **Browse**, **Browse logs…** and **Browse recovery files…** provide file trees with filename filtering, copy export, and access to the native folder; completed CSV logs open in the built-in viewer.

Open in > Tuning Suite requires an installed compatible BimmerStein Tuning Suite and is disabled when it is absent or needs an update. Launch rechecks the installation and validates the selected Bin. It creates an editable copy in `editor/`, keeping the catalogued original unchanged. Import the result after saving in Tuning Suite.

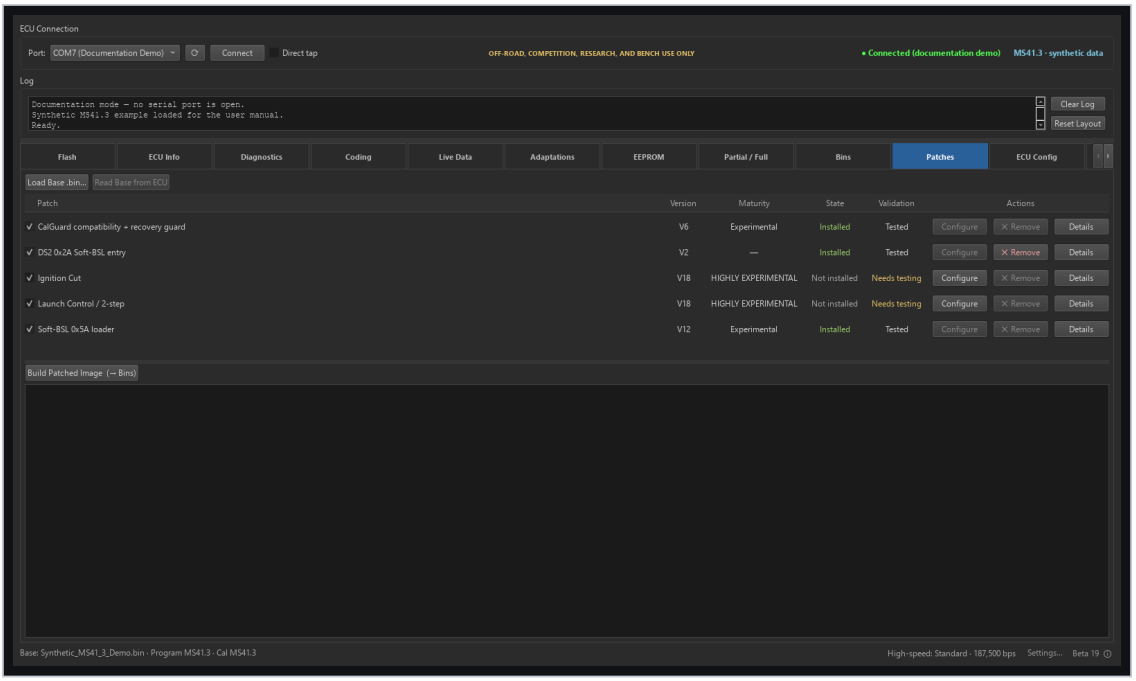
Select exactly two entries and choose **Compare** for a read-only report of their SHA-256 identity, program/calibration variants, checksum state, installed patches, changed-byte count, and changed address ranges. Matching full and calibration files compare their calibration regions using the correct family layout. Compare does not change either file. Older entries receive a baseline hash when first loaded after their stored size is checked; that baseline cannot prove the file's earlier history.

For two EEPROM entries, Compare shows decoded field and raw-byte differences when their known layouts match. Unknown or differing layouts are compared as raw bytes only.

Select a 256 KB full-ROM entry and choose **Patches** to load it directly into the Patches tab. Adding or removing patches works on an in-memory copy; **Build Patched Image** archives a new Bin and never overwrites the selected original.

Select a 24 KB tune or 256 KB full-ROM entry and choose **BSL-Unbricker** to load it as the hardware-BSL reference image and open that tab. This prepares the recovery controls only; it does not connect to hardware, review or approve a flash plan, or flash the ECU. A tune selects the **tune** region when available. A full ROM leaves the chip, physical half, and region unchanged for the operator to select explicitly.

Firmware patches



Patch selection with maturity, installation and validation status

The Patches tab loads a compatible base image, checks which patches apply, detects already-installed and deprecated revisions, validates dependencies and byte collisions, recomputes required checksums, and archives the composed image into Bins.

The footer shows the base filename and program/calibration identity; hover for the full path and details. Operation progress temporarily takes priority. Hybrid warnings stay at the start of the summary.

Select an editable patch and choose **Configure**. **Configure *** marks pending settings; **Build Patched Image** saves them with the patches and makes the saved image the working base. No intermediate BIN or ECU readback is needed. Deselecting a patch discards its pending settings; loading another base clears all pending changes.

Numeric settings use number controls. You can type an in-range value; **Review Changes** shows its rounded build value. The selected **Follow stock**, **Automatic**, or **Legacy zero** button preserves that mode. Follow stock shows the loaded calibration's value when it is a fixed scalar; injection and original VANOS logic are calculated by the ECU. Toggle the mode off to enter a custom value.

The list shows each patch's version, maturity, installation state, validation and blocking warnings. **Details** adds the description, ID, dependencies and boot-write requirements; press **Escape** or click outside to close it. **Configure**, **Remove** and **Details** stay in the same position in every row; unavailable actions are greyed out.

Status and migration

- **Installed / Not installed** identifies whether that revision's signature is present in the loaded image.
- **Maturity** records the patch's stage, including Experimental, Beta, Stable or Deprecated. Ignition Cut and Launch Control show **HIGHLY EXPERIMENTAL**; **-** means no maturity was assigned.
- **Deprecated - remove only** identifies a historical revision retained for safe detection and removal. Deprecated descriptors are not offered for a new installation.
- **Untested / Needs testing** marks incomplete vehicle testing or required physical validation. **Tested** records confirmed testing within the scope in **Details**.
- **Emulator verified**, **Bench tested**, and **Vehicle tested** identify the recorded test scope. Emulator or bench results do not imply vehicle validation; **Details** retains the full proof notes.
- Boot-region patches require a transfer path that can actually deliver their bytes.

Ignition Cut V18 and Launch Control V18 are current for MS41.0, MS41.1, MS41.2 and MS41.3. Launch Control requires the matching ignition-cut revision, but the standalone Ignition Cut switch may remain Off. They retain compatible configured controls during upgrade and include the later timing and injection-recovery corrections. These revisions are **HIGHLY EXPERIMENTAL** and require vehicle testing. Offline verification does not establish safe engine behavior. AlphaN MAF-failsafe V3 is **Tested**. AMD 29F Flash Driver V5, CalGuard V6, Soft-BSL entry V2 and loader V12 have **Tested** validation status following owner confirmation. This status applies to the recorded configurations; check **Details** for the test scope. Ignition Cut and Launch Control V18 still require testing. Bench results are specific to the tested ECU, flash device and bank; **Details** records that scope. Removing CalGuard also removes its startup wait while preserving Soft-BSL and the flash driver.

Upgrading existing patches

Compatible, exactly recognized predecessors can upgrade to the matching current ignition/launch pair while retaining their configured settings. Review both configurations, new controls and dependencies before flashing. A partial or mismatched installation must follow the application's removal or recovery instructions; do not combine incompatible revisions.

Deprecated revisions remain detectable for upgrade or removal. Historical instructions are not the installation procedure for the current pair. Updating the application alone does not change firmware already installed in an ECU.

Affected learning and observations wait for native recovery conditions and fresh samples. Electrical faults, fallback behavior, DTC100/DTC214, prior misfire or shutdown evidence and completed catalyst results remain effective. A DTC number alone does not identify its cause.

Applying an experimental patch requires explicit confirmation. Offline validation does not prove safe behavior on an engine.

> **DANGER: IGNITION CUT HAZARD.** Ignition Cut remains experimental. It may suppress spark while injection continues at the stock or configured fixed pulse width. Unburned fuel can damage catalytic converters and exhaust components; never use it on a car with catalytic converters. Its fuel-adaptation and diagnostic guards are offline exact-byte verified but not vehicle-validated.

Bundled patch definition

The release includes `BimmerStein MS41 Patch Definitions.xml` beside the executable. Load it into a compatible calibration editor to configure items added by the matching patches. Install the firmware patch first and verify the ECU variant, calibration ID, and patch revision. A mismatched definition can expose incorrect tables or write to the wrong calibration addresses.

Launch arming and release

Always (no switch) treats the input as active while retaining the speed and throttle gates. It arms only below Arm Speed with TPS at or above Min TPS. An armed latch remains active during rollout until speed reaches Max Speed or TPS drops below Min TPS; releasing the clutch preserves this latch for the rollout. Releasing a physical switch ends optional fixed timing even while the limiter latch remains active. After release, it can rearm only below Arm Speed. Arm Speed must be at least 1 km/h; set it to 1 to allow arming at a reported speed of zero. Clutch polarity applies only to physical switch modes. A saved zero Arm Speed in an older configuration must be corrected before enabling or updating an enabled setup.

Fuel-cut launch mode

Fuel mode uses the lower of the launch hard setting and the native hard threshold. Automatic still requests soft cut + 96 RPM, capped at that native threshold. Soft RPM independently uses the lower of native soft and launch soft. Hard RPM may be below, equal to, or above soft RPM; setting it lower does not change soft RPM. The hard path accelerates the native staged cut rather than instantly cutting every cylinder. Stock behavior can begin restoring fuel while RPM is still above a lower hard setting.

Two fuel-only settings default to **Follow stock**:

- **Fuel hysteresis B:** staged restoration begins strictly below effective soft RPM minus this value.
- **Fuel hysteresis A:** clears remaining limiter stages strictly below effective soft RPM minus this value. Staged restoration may finish before this threshold is crossed.

Both settings use 32 RPM steps, accept 0-8128 RPM, and remain independent. Subtraction saturates at zero. **Follow stock** reads the current corresponding stock calibration. When stock soft governs, either stock RPM threshold is reached, or launch disengages during an active cut, stock hysteresis remains selected until that cut episode ends. Rearming during that episode does not reset the native counters or replace stock hysteresis. The ignition-only settings are ignored in fuel mode.

Ignition-cut mode and fueling

In launch ignition mode, hard RPM is ignored. Spark cut starts at or above launch soft RPM and releases strictly below soft RPM minus Ignition RPM hysteresis, unless the speed or throttle gates release launch first. All RPM values use 32 RPM steps. The current MS41.3 Launch Control uses dedicated calibration storage and can be configured with boost control. Follow any removal prompt for a historical installation that overlaps other calibration data.

Standalone Ignition Cut and Launch ignition mode have separate RPM hysteresis and fixed injector-pulse-width settings. Fixed width applies while the respective spark-cut request is active; selecting an ignition retard range does not enable it. **Preserve stock injection** uses the ECU's calculated injection. A custom **0 ms** is a valid zero base pulse, not a guaranteed injector shutoff. Native cylinder and battery corrections, scheduling and fuel shutoffs still apply. If the corrected pulse cannot fit the validated timing interval, V18 returns to native fueling. Launch fuel mode ignores these ignition-only settings and uses the native staged fuel limiter.

Optional ignition timing

Standalone Ignition Cut and Launch ignition mode each have **Ignition retard range** and **Ignition-cut fixed angle** settings. The range defines an RPM region below the corresponding spark-cut threshold; it is separate from RPM hysteresis and does not set fueling. Its initial value is **192 RPM**. An enabled angle requires a positive range smaller than the cut RPM.

Ignition-cut fixed angle defaults to **Off**. A custom angle is absolute, from -23.625 to 0 degrees BTDC in 0.375-degree steps. Negative values mean after top dead center. Zero selects top dead center; it is not the same as Off.

Launch fuel mode has its own **Fuel-cut ignition angle**, default **Off**, with the same absolute-angle range and steps described above. The override begins when an eligible fuel-mode launch reaches its first soft or hard threshold. It continues through fuel restoration until all fuel returns and RPM is below that threshold. Releasing the selected physical switch, leaving the speed/throttle gates or native limiter intervention ends the optional fuel-mode timing request. Arming alone does not change timing. Ignition-cut mode ignores this fuel-mode setting; normal dwell and injection remain under their existing controls.

For active optional timing, V18 uses the later of the selected angle and the ECU's native protected timing: it does not advance timing past that protection. Native input and knock-sensor faults retain their veto. Learned knock corrections are preserved; affected learning waits through actual spark cuts or timing changes and the required fresh-sample recovery.

Ordinary requested-angle telemetry does not show this final scheduling override. Software checks do not replace bench and vehicle validation.

Safe composition

1. Load or read a compatible base image.
2. Remove any detected predecessor when instructed.
3. Select the required current patches and dependencies, then use **Configure** to set their parameters.
4. Review Untested and blocking warnings, and check boot-write requirements in **Details**.
5. Build the image and inspect the build log.
6. Flash the archived result through the normal Flash workflow only after reviewing it.

ECU Config and VIN / EWS

ECU Config

MS42/MS43 offers revision-specific calibration and supported program controls. Read the connected ECU or open a matching file before editing. Program changes use the shared full-image writer; calibration-only changes use the calibration route. Immobilizer and checksum options appear only for supported revisions. No global VANOS, complete knock-disable, IACV conversion or complete ASC control is assumed from a similarly named setting. Octane-adaptation learning does not disable knock detection or correction.

The following details describe MS41:

ECU Config exposes supported calibration configuration bits in either file mode or connected-ECU mode. Read the current values first, change only understood options, and write through the normal guarded path. Some options affect checksum verification or core engine features; their warnings are not interchangeable with the optional host read-back Verify checkbox.

Oxygen-feedback choices are resolved from the exact CAL ID because ID41, ID42, ID59, and ID85 do not share one Byte 6 encoding. The ID12/ID60 O2 disable remains experimental and is available only for a full-ROM target. Select **O2 Feedback Program Gate = Feedback Disabled** first; the calibration section will then expose **Oxygen Sensors = Disabled (Experimental)**. Switching the program gate back to enabled removes that calibration choice.

In connected-ECU mode, **Read from ECU** enables the supported program controls. If the current connection already has an archived full read, the application reuses it. Otherwise, a program change makes **Write to ECU** read and archive the ECU's unmodified 256 KB ROM before applying the selected configuration. Calibration-only changes retain the faster 24 KB partial-write route. Program changes are passed to the same guarded full-ROM writer used by the Flash tab, including automatic Soft-BSL/native-fast/standard-DS2 selection and fallback, checksum and flash-family checks, boot-region policy, confirmation, and optional read-back verification. The configuration workflow never uses a different-variant base image as a firmware conversion.

VIN editing

For MS42/MS43, **Read VIN / AIF** reads the current VIN and programming history. VIN writes use a free history slot where possible and handle full or interrupted history through the shared RAM write/recovery workflow. Keep original backups.

Virginize ECU clears supported ISN/pairing data. **Also clear VIN and programming history** is optional. EWS alignment is not performed automatically. The altered pairing behavior of 43X001 is not offered stock virginizing.

For MS41, VIN editing is a Soft-BSL sector read-modify-write workflow. It requires a compatible installed loader, detected flash family, D2XX connection, and a live identity-sector cache tied to the current connection. The application changes only the packed VIN field and preserves the rest of the owning erase sector.

MS41 EWS2 alignment

EWS alignment is separate from VIN editing. It reads a fresh live DME ISN, applies the validated EWS2 encoding, rechecks the value immediately before transmission, and requires the expected EWS acknowledgement.

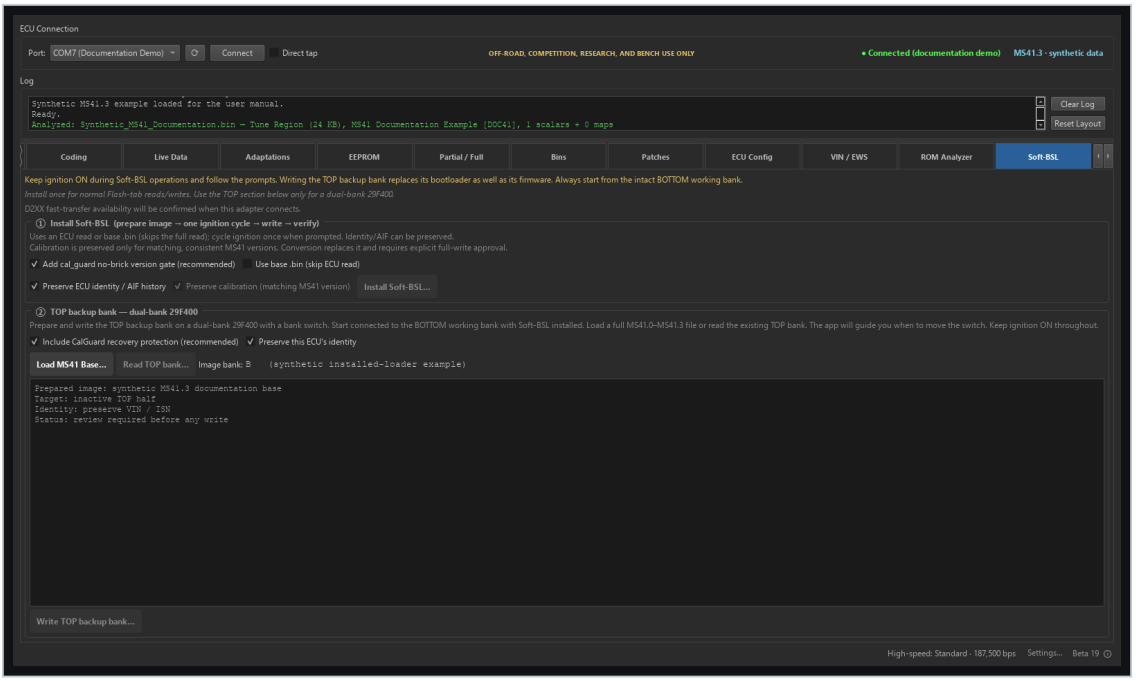
MS42/MS43 EWS synchronization

MS42/MS43 provides **Synchronize EWS...** and **Reset Start Value...** for its EWS3.3 workflow. Both use the connected DME and EWS module and require explicit confirmation. Read the returned result: a mismatch or an unavailable start value is not successful synchronization.

These are separate from MS41's EWS2 ISN-transfer procedure, VIN editing and virginizing. Virginizing does not automatically run either action; pairing remains a separate user choice.

> **WARNING:** Do not assume that a cached VIN/identity read proves the current EWS state. VIN and EWS use separate live workflows and separate ownership checks.

Soft-BSL - MS41



Soft-BSL installation and recovery controls

Soft-BSL installs a small persistent loader in ECU flash. The loader accepts an agent over DS2, loads it into RAM, and transfers control to that agent for higher-speed reads and writes.

- Full and tune reads use the same optimized RAM-agent path.
- Intel and AMD/JEDEC devices use their matching agent and command set.
- Program writes on MS41.2 include the checksum storage required by its enabled program CRC.
- Successful writes finalize the marker independently of optional host read-back verification.

Installation

The guided installer identifies the ECU, validates its image and flash family, prepares temporary entry, installs the persistent loader and confirms normal DS2 after the required ignition sequence. It removes the temporary entry afterward; daily operations use the persistent loader and current RAM agents.

If the `E659=0xCC` readiness check times out, the application releases the adapter and asks for an ignition cycle followed by **Retry** or **Cancel**. No authorization, erase or programming request has been sent. Retry opens a fresh native-fast session; each repeated timeout requires a new decision.

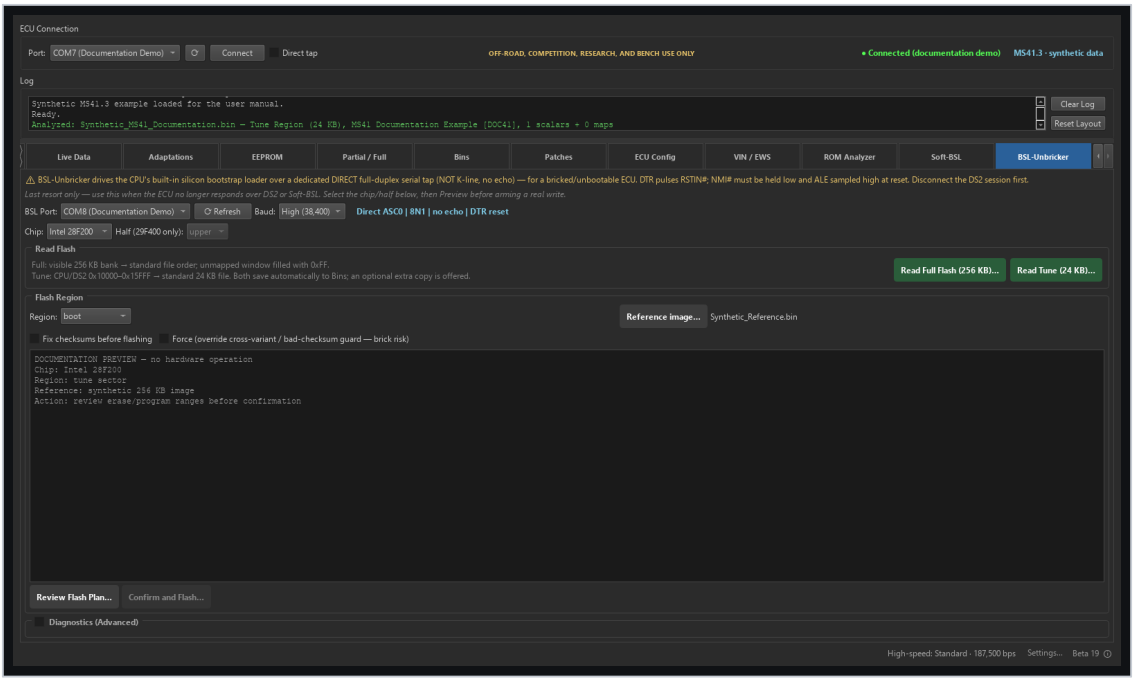
Cross-bank and boot writes

Dual-bank support for the 29F400BB is tested and working. The tool can write both halves and manage each bank's patches and Soft-BSL.

New AMD TOP images automatically include a resident DS2 guard. It rejects full/program writes before erase while allowing normal 24 KB calibration writes. Soft-BSL full and partial writes remain available through the existing workflow. BOTTOM images retain normal DS2 behavior. The guard requires installing the prepared TOP image; an application update alone does not modify the ECU. Installation, cold recovery and normal startup were tested on one MS41.0 / AM29F400BB bench ECU. The DS2 guard has offline coverage; destructive stock-DS2 writes against it were not tested on hardware.

For advanced golden-bank or boot writes, follow the displayed A17/bank instructions and verify the physical half. Keep normal tune/program writes on their usual route.

BSL-Unbricker - MS41



Hardware BSL review and programming controls

Hardware BSL communicates through the 80C166 bootstrap loader even when flash contents are blank or corrupt.

Required connection

- Separate full-duplex ASC0 Tx/Rx direct tap.
- 8N1 serial framing with no K-Line echo removal.
- DTR-controlled reset pulse when wired to the supported reset circuit.
- Correct ALE/NMI bootstrap-entry straps.
- Correct chip selection and 29F400 half.

Selectable rates are 9,600, 19,200, and 38,400 baud. Use 38,400 as the preferred D2XX rate and select 19,200 or 9,600 when adapter, wiring, or signal stability requires a fallback.

Intel VPP

Intel 28F200 erase/program requires the correct 12 V VPP/RP# supply. AMD/JEDEC parts do not. The VPP control remains disabled until the Intel chip family is selected.

Read and write workflow

1. Select the dedicated BSL COM port, chip, physical half, region, and rate. The application uses the DTR reset line for the supported entry circuit.
2. For a read, choose full 256 KB file order or the standard 24 KB tune.
3. For a write, select the reference image directly or prepare one from **Bins**, then choose **Review Flash Plan**.
4. Verify every physical address range and prerequisite in the preview.
5. Choose **Confirm and Flash** only when the plan is correct.
6. Monitor erase, programming, and complete physical-region read-back.
7. Remove VPP and bootstrap straps before returning to normal K-Line operation.

Troubleshooting

The COM port is unavailable

- Close other software using the adapter.
- Refresh the port list.
- Confirm whether the normal K-Line adapter or the separate BSL adapter is required.
- Reconnect the FTDI device and verify its Windows driver.

D2XX is unavailable

- Confirm the selected COM port belongs to the intended FTDI device.
- Verify the FTDI D2XX driver is installed.
- Normal DS2 can use the standard serial fallback at 9600.
- Hardware BSL can use its compatible serial fallback. Native-fast DS2 requires D2XX; Soft-BSL can fall back to its supported low-rate serial tier when D2XX is unavailable.

The fast-path stability check fails

If a Soft-BSL tier fails before erase, the application can retry a lower Soft-BSL tier. If a native-fast DS2 check fails before erase and normal ECU state is confirmed, it can restart the complete operation over normal DS2. Check adapter latency, wiring, ground, ECU voltage, and signal integrity before retrying.

A write failed

- If the UI says erase did not start, correct the reported connection or file problem and retry.
- If the UI says recovery is active, keep ignition ON and use **Retry Flash Recovery**.
- If an MS41 no longer boots and no retained session exists, prepare its hardware-BSL connection.
- For MS42/MS43 in that state, use a suitable separate hardware recovery method; the app's BSL-Unbricker is MS41-specific.

The ROM Analyzer cannot load definitions

Use **Load Definition...** to select a matching MS41 XML or MS42/MS43 XDF. Do not copy definitions into `_internal` or any other packaged runtime directory. If a registered definition was changed or damaged outside the application, delete it and import a known-good copy again. Confirm the BIN size and exact ECU software identity before relying on matched values.

Values or identity look wrong

Stop the operation. Confirm the connected ECU, file, variant, byte order, and definition. Re-read the value through an independent path before writing anything.

Exporting a support bundle

Choose the build label at the lower-right of the main window, then **Export Support Bundle**. The ZIP contains build information plus, when available, privacy-scoped metadata and SHA-256 for the single selected Bin, the latest Live Data CSV, and the latest native-fast journal. Raw ROM bytes are never included.

A session log is optional because it may contain VIN, ISN, ECU identifiers, local paths, and detailed errors. The application asks before including it. Bin filenames, VIN, ISN, notes, and ECU identifiers are excluded from the default metadata, and generic ZIP member names are used.

Final checklists

Before a normal read

- [ ] ECU voltage is stable.
- [ ] The normal K-Line/direct-tap selection matches the wiring.
- [ ] The correct COM port is selected.
- [ ] Full ROM or calibration is intentionally selected for the connected ECU family.
- [ ] The output location does not overwrite an irreplaceable file.

Before a normal write

- [ ] The target file size and ECU variant match the intended job.
- [ ] The detected flash-chip family is compatible with the image and path.
- [ ] Checksum correction is enabled unless there is a documented reason otherwise.
- [ ] Backup and Verify choices reflect the operator's decision.
- [ ] Stable power and hardware recovery are available.
- [ ] No other software owns the COM port.

Before MS41 hardware BSL programming

- [ ] Direct ASC0 wiring and bootstrap straps are correct.
- [ ] Chip family and 29F400 half are correct.
- [ ] Intel VPP is active only for the Intel write phase.
- [ ] The reviewed physical address plan matches the intended region.
- [ ] The reference image is the correct size and byte order.
- [ ] A recovery plan exists if programming is interrupted.

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